Uncertainty Quantification Using Multi-Dimensional Hermite Polynomials

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Abstract

The general stochastic problem involves the propagation of input uncertainties through a computation model to arrive at a random output vector. This paper presents the application of the multi-dimensional Hermite polynomials to reduce an unknown random output vector into a significantly simpler unknown vector of numbers. The unknown numbers are evaluated using a collocation method because it has the important practical advantage of allowing existing deterministic numerical codes to be used as "black boxes". A simple laterally loaded pile example involving two input random variables demonstrated that a third- or fourth-order Hermite expansion is adequate to reproduce probabilities of failure between 10^{-3} and 10^{-4}. A simple and efficient 2-term recurrence method for obtaining Hermite polynomials of any order in the case of $n = 2$ is proposed. To our knowledge, this proposal appears to be original.

Introduction

The efficient propagation of input uncertainties through a physical model to arrive at a vector of output uncertainties is an outstanding challenge. Ideally, the full finite-dimensional distribution function of the random output vector is desired, although second-moment characterization may be sufficient in some applications. In principle, Monte Carlo simulation can be used to solve this problem, regardless of the complexities underlying the physical model and/or input uncertainties. One may assume with minimal loss of generality that a complex geotechnical problem would only admit numerical solutions and the spatial domain can be modeled by a

scalar/vector random field. Monte Carlo simulation requires a procedure to generate realizations of the input uncertainties and a numerical scheme for computing a vector of outputs from each realization. The former step is not necessarily trivial and not completely solved even in the theoretical sense. Details are outside the scope of this paper and given elsewhere (Phoon, 2006). The latter step is identical to the repeated application of a deterministic solution process. The only potential complication is that a particular set of input parameters may be too extreme, say producing a near-collapse condition, and the numerical scheme may become unstable. The statistics of the random output vector are contained in the resulting ensemble of numerical output values produced by repeated deterministic runs. Fenton and Griffiths have been applying Monte Carlo simulation to solve soil-interaction problems within the context of a random field since the early nineties (e.g., Griffiths & Fenton, 1993, 1997, 2001; Fenton & Griffiths, 1997, 2002, 2003).

For a sufficiently large and complex soil-structure interaction problem, it is computationally intensive to complete even a single run. The rule-of-thumb for Monte Carlo simulation is that $10/p_f$ runs are needed to estimate a probability of failure, p_f, within a coefficient of variation of 30% (Au, 2001). The typical p_f for a geotechnical design is smaller than one in a thousand and it is expensive to run a numerical code more than ten thousand times even for a modest size problem. This significant practical disadvantage is well known. A variety of variance reduction techniques such as importance sampling, Markov chain Monte Carlo simulation, subset simulation (Au, 2001), and quasi-Monte Carlo simulation methods (Robinson & Atcitty, 1999) have been proposed to reduce the number of runs without compromising accuracy. An alternate strategy is to replace the expensive numerical model with an approximate but cheaper surrogate version. The most common method is the response surface model (Box & Draper, 1987) where second- or third-order polynomial functions are often used as surrogates (Wong, 1985). Another relatively cheap and perhaps more general surrogate model is the neural network (Goh & Kulhawy, 2005).

A promising approach that is gaining wider attention is the spectral stochastic finite element method originally proposed by Ghanem & Spanos (1991). The key element of this approach is the expansion of the unknown random output vector using multi-dimensional Hermite polynomials as basis functions (also called a polynomial chaos expansion). The unknown deterministic coefficients in the expansion can be solved using the Galerkin or collocation method. The former method requires significant modification of the existing deterministic numerical code and is impossible to apply for most engineers with no access to the source code of their commercial softwares. Sudret & Der Kiureghian (2000) provided implementation details for the Galerkin method. The collocation method can be implemented using a small number of fairly simple computational steps and does not require the modification of existing deterministic numerical code. The objective of this paper is to present the stochastic collocation method with emphasis on computation rather than theoretical details. More technical exposition of the polynomial chaos expansion is given elsewhere (e.g., Ghanem & Spanos, 1991). The potential of this method for uncertainty quantification is illustrated using a simple laterally loaded pile example.

Hermite Polynomials

One-dimensional expansion. One-dimensional Hermite polynomials are given by:

$$\begin{aligned} &H_0(U) = 1 \\ &H_1(U) = U \\ &H_2(U) = U^2 - 1 \\ &H_3(U) = U^3 - 3U \\ &H_{k+1}(U) = U\,H_k(U) - kH_{k-1}(U) \end{aligned} \tag{1}$$

where U is a standard Gaussian random variable (mean = 0 and variance = 1). It can be proven rigorously (Phoon, 2003) that any random variable X (with finite variance) can be expanded as follows:

$$X = \sum_{k=0}^{\infty} a_k H_k(U) \tag{2}$$

The numerical values of the coefficients, a_k, depend on the distribution of X. The key practical advantage of Eq. (2) is that the randomness of X is completely accounted for by the randomness of U, which is a <u>known</u> random variable.

Phoon & Honjo (2005) demonstrated the usefulness of Eq. (2) using a gamma random variable with probability density function defined by:

$$q(x) = \frac{1}{b^c \Gamma(c)} x^{c-1} e^{-\frac{x}{b}} \tag{3}$$

where b is the scale parameter and c is the shape parameter. Their results based on a simulation sample size = 50000 are reproduced in Fig. 1 for the cases of b = 1, c = 0.5 (top) and b = 1, c = 2 (bottom). The former distribution is related to a chi-square distribution with one degree of freedom. Although it is highly skewed, a 6-term Hermite expansion is sufficient to match probabilities as low as 10^{-4}. The second gamma distribution looks "Gausian" and it is not surprising that an even shorter 4-term Hermite expansion is enough for this case.

It is possible to generalize Eq. (2) for a random vector $\underline{X} = (X_1, X_2, \ldots X_n)'$ as follows:

$$X_i = \sum_{k=0}^{\infty} a_{ik} H_k(U_i) \tag{4}$$

The key step is to select an appropriate covariance matrix for the n-dimensional Gaussian vector, $\underline{U} = (U_1, U_2, \ldots U_n)'$, such that the desired covariance matrix of $\underline{X}$ is achieved using:

$$\mathrm{E}X_i X_j = \sum_{k=0}^{\infty} k!\, a_{ik} a_{jk}\, \rho_{Ui,Uj}^{k} \tag{5}$$

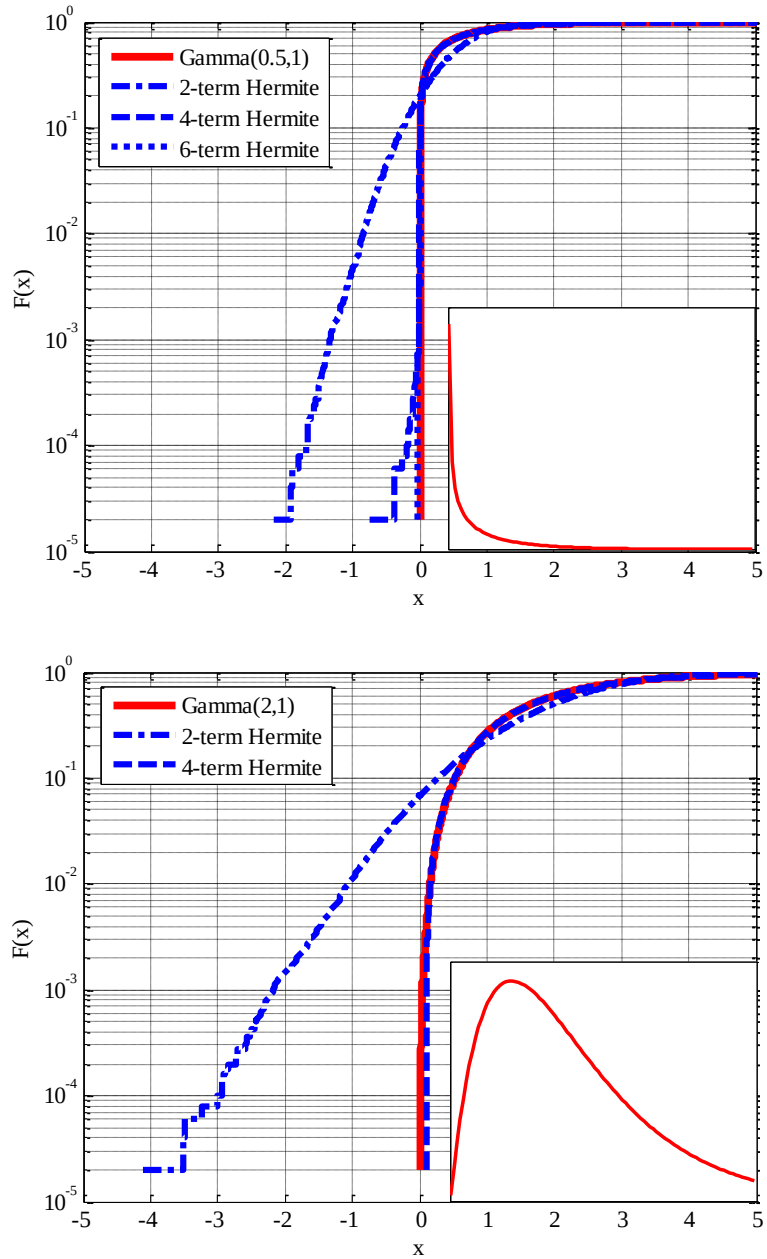


Figure 1. Cumulative distribution function of Hermite expansion for two gamma distributions. (Source: Phoon & Honjo, 2005)

where ρ_{UiUj} is the correlation between U_i and U_j. Examples are given by Phoon (2006). It suffices to note that Eq. (5) may not produce a valid covariance matrix for $\underline{U}$ and this limitation is fundamental. If $\underline{X}$ is an output vector from a numerical code, the covariance of $\underline{X}$ is unknown and hence the Gaussian random vector $\underline{U}$ underlying Eq. (4) is also unknown. The bottomline is that a general random vector $\underline{X}$ expanded in the form of Eq. (4) cannot enjoy the same advantage as Eq. (2), where the only unknowns are the coefficients of the series. This practical problem is solved by using multi-dimensional Hermite polynomials, which are supported by a <u>known</u> Gaussian

random vector with zero mean, unit variance, and uncorrelated or independent components (called a standard Gaussian vector).

Multi-dimensional expansion. The multi-dimensional Hermite polynomials are significantly more complex than the one-dimensional version. Two approximate second- and third-order forms are respectively given below (Isukapalli, 1999):

$$X \approx a_{\mathrm{o}} + \sum_{i=1}^{n} a_i U_i + \sum_{i=1}^{n} a_{ii}\left(U_i^2 - 1\right) + \sum_{i=1}^{n-1}\sum_{j>i}^{n} a_{ij} U_i U_j \tag{6a}$$

$$\begin{aligned} X \approx\ & a_{\mathrm{o}} + \sum_{i=1}^{n} a_i U_i + \sum_{i=1}^{n} a_{ii}\left(U_i^2 - 1\right) + \sum_{i=1}^{n} a_{iii}\left(U_i^3 - 3U_i\right) + \sum_{i=1}^{n-1}\sum_{j>i}^{n} a_{ij} U_i U_j + \\ & \sum_{i=1}^{n}\sum_{\substack{j=1 \\ j\neq i}}^{n} a_{ijj}\left(U_i U_j^2 - U_i\right) + \sum_{i=1}^{n-2}\sum_{j>i}^{n-1}\sum_{k>j}^{n} a_{ijk} U_i U_j U_k \end{aligned} \tag{6b}$$

For n = 3, Eqs. (6a) and (6b) produce the following expansions (indices for coefficients are re-labeld consecutively for convenience):

$$\begin{aligned} X \approx\ & a_{\mathrm{o}} + a_1 U_1 + a_2 U_2 + a_3 U_3 + a_4\left(U_1^2 - 1\right) + a_5\left(U_2^2 - 1\right) + a_6\left(U_3^2 - 1\right) + \\ & a_7 U_1 U_2 + a_8 U_1 U_3 + a_9 U_2 U_3 \end{aligned} \tag{7a}$$

$$\begin{aligned} X \approx\ & a_{\mathrm{o}} + a_1 U_1 + a_2 U_2 + a_3 U_3 + a_4\left(U_1^2 - 1\right) + a_5\left(U_2^2 - 1\right) + a_6\left(U_3^2 - 1\right) + \\ & a_7 U_1 U_2 + a_8 U_1 U_3 + a_9 U_2 U_3 + a_{10}\left(U_1^3 - 3U_1\right) + a_{11}\left(U_2^3 - 3U_2\right) + \\ & a_{12}\left(U_3^3 - 3U_3\right) + a_{13}\left(U_1 U_2^2 - U_1\right) + a_{14}\left(U_1 U_3^2 - U_1\right) + a_{15}\left(U_2 U_1^2 - U_2\right) \\ & + a_{16}\left(U_2 U_3^2 - U_2\right) + a_{17}\left(U_3 U_1^2 - U_3\right) + a_{18}\left(U_3 U_2^2 - U_3\right) + a_{19} U_1 U_2 U_3 \end{aligned} \tag{7b}$$

In general, N_2 and N_3 terms are respectively required for the second-order (Eqs. 6a) and third-order (Eqs. 6b) expansions (Isukapalli, 1999):

$$\begin{aligned} N_2 &= 1 + 2n + \frac{n(n-1)}{2} \\ N_3 &= 1 + 3n + \frac{3n(n-1)}{2} + \frac{n(n-1)(n-2)}{6} \end{aligned} \tag{8}$$

For a fairly modest random dimension of n = 5, N_2 and N_3 terms are respectively equal to 21 and 56. Hence, fairly tedious algebraic expressions are incurred even at a third-order truncation. Multi-dimensional Hermite polynomials are mostly computed using symbolic algebra at present. The accuracy of these truncated low-order expansions are discussed elsewhere (Field & Grigoriu, 2004; Debusschere et al. 2004).

Stochastic Collocation Method

Algorithm. The random output (assumed to be scalar for simplicity) from a numerical code can be expressed as an implicit function of random input parameters, i.e. $Y = \mathrm{f}(X_1, X_2, \ldots X_\mathrm{n})$. The random input parameters $(X_1, X_2, \ldots X_\mathrm{n})$ representing load, soil, and/or geometrical parameters, can be correlated and non-Gaussian. As a result, the output Y is usually non-Gaussian, following an unknown probability distribution function. If the output is a vector containing say two components (Y_1, Y_2), the unknown to be computed is a bivariate probability distribution function. The stochastic collocation method can be summarized as follows:

1. If $(X_1, X_2, \ldots X_\mathrm{n})$ constitutes a correlated Gaussian vector, it is possible to express this random input vector using a standard Gaussian vector $(U_1, U_2, \ldots U_\mathrm{n})$ via Cholesky or spectral decomposition (Phoon et al., 2004). The spectral method can potentially reduce the size of the input random dimension below n by discarding small eigenmodes and is preferred because the number of terms in Hermite expansions proliferates rapidly with n (Eq. 7). A Gaussian random field can be converted to a discrete standard Gaussian vector using the Karhunen-Love expansion (Phoon et al., 2002). Non-Gaussian random vectors can be constructed by applying the appropriate cumulative distribution functions to each component of a suitably correlated Gaussian vector (Phoon, 2006). Phoon (2004) showed that this translation method is identical to Eq. (4). A non-Gaussian random field can be constructed in a similar way.
2. Thus, the output Y can be expressed as $\mathrm{f}(U_1, U_2, \ldots U_\mathrm{n})$ or possibly with fewer arguments if small eigenmodes are discarded. The random variable Y with unknown distribution can also be expressed using one of the approximations given in Eq. (6). For $n = 2$ and selecting a third-order approximation, we can write:

$$\begin{aligned}\mathrm{f}(u_{i1}, u_{i2}) &= a_\mathrm{o} + a_1 u_{i1} + a_2 u_{i2} + a_3\left(u_{i1}^2 - 1\right) + a_4\left(u_{i2}^2 - 1\right) + a_5 u_{i1} u_{i2} + \\ &\quad a_6\left(u_{i1}^3 - 3u_{i1}\right) + a_7\left(u_{i2}^3 - 3u_{i2}\right) + a_8\left(u_{i1} u_{i2}^2 - u_{i1}\right) + a_9\left(u_{i2} u_{i1}^2 - u_{i2}\right) \\ &= a_\mathrm{o} + a_1 n_{i1} + a_2 n_{i2} + \cdots + a_9 n_{i9}\end{aligned} \tag{9}$$

where (u_{i1}, u_{i2}) is one possible realization from a standard Gaussian vector (U_1, U_2), $(1, n_{i1}, n_{i2}, \ldots, n_{i9})$ are Hermite basis functions evaluated at (u_{i1}, u_{i2}), and $(a_0, a_1, a_2, \ldots, a_9)$ are unknown but constant numbers. The unknown numbers can be determined using 10 different realizations. In matrix notation, we have:

$$N\underline{a} = \underline{f} \tag{10}$$

where N is a 10× 10 matrix with ith row given by $(1, n_{i1}, n_{i2}, \ldots, n_{i9})$, f is a 10× 1 vector with ith component given by $\mathrm{f}(u_{i1}, u_{i2})$ and $\underline{a}$ is a 10× 1 vector containing unknown numbers $(a_0, a_1, a_2, \ldots, a_9)'$. This is known as the stochastic collocation method. It is also possible to solve for the unknown coefficients using regression if there are more than 10 realizations.

$$N'N\underline{a} = N'\underline{f} \tag{11}$$

where N is a $m \times 10$ matrix and m is the number of realizations.

3. Tatang (1995) proposed the placement of collocation points at the roots of the next higher order Hermite polynomial. For a second-order expansion, the roots of the third-order Hermite polynomial are 0 and $\pm\sqrt{3}$ [see fourth row of Eq. (1)]. For a third-order expansion, the roots of the fourth-order Hermite polynomial are $\pm\sqrt{(3 \pm\sqrt{6})}$. Although zero is not one of the roots, it should be included.

Example. The example considered is a laterally-loaded free-head rigid pile in sand (Fig. 2). The performance function (G) is given by:

$$G = FS - 1 = 0.5M \frac{\gamma BD^3 \tan^2\left(45^\circ + \phi/2\right)}{\left(e + D\right)F} - 1 \tag{12}$$

where M is the random factor describing the model uncertainty, B and D are respectively the diameter and length of the pile, e is the eccentricity, γ is the unit weight of sand, ϕ is the effective friction angle, and F is the applied load. The model factor M can be adequately modeled as a log-normal random variable with mean = 1.3 and standard deviation = 0.5. The applied load F is assumed to be lognormally distributed with mean = 1500 kN and standard deviation = 300 kN. The rest of the parameters are assumed to be deterministic and were given by: D = 10 m, B = 1 m, e = 1 m, γ = 18 kN/m^3 and ϕ = 40°. The mean factor of safety is 3.3. For this simple example, the reliability index can be computed exactly because only two lognormal random variables are involved. The exact reliability index is 2.69 and the probability of failure is 0.0036.

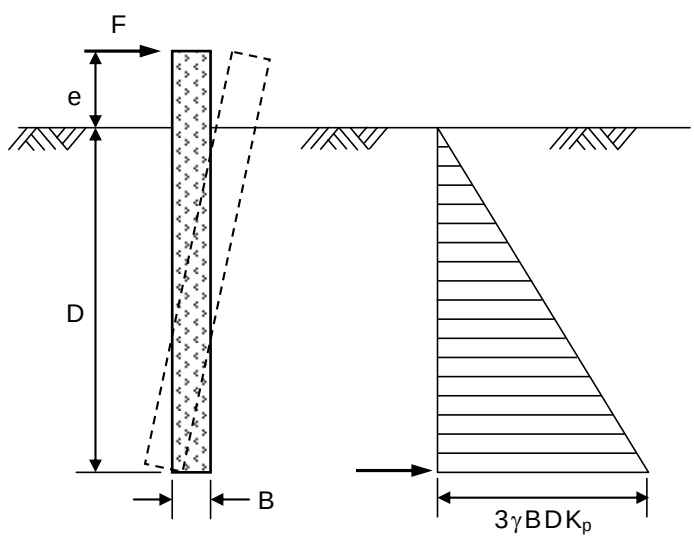


Figure 2. Simplified Broms approach for laterally-loaded free-head rigid pile in sand.

It is easy to express the random input parameters in terms of standard Gaussian random variables: $M = \exp(0.371U_1+0.193)$ and $F = \exp(0.198U_2+7.294)$. The performance function can be approximated as a third-order Hermite expansion containing U_1 and U_2. From Eq. (9), it is clear there are 10 unknown coefficients. Given that there are 5 collocation points for each random variable [0 and $\pm\sqrt{(3 \pm\sqrt{6})}$], there are 25 possible realizations for (U_1, U_2). The coefficients can be computed from these realizations using regression, Eq. (11).

Fig. 3(top) compares the cumulative distribution function of the performance function determined from direct simulation of Eq. (12) and from Hermite expansion. The simulation sample size is 500,000. The probability of failure from a third-order expansion is 0.0033 (direct simulation is 0.0035). A second example is analyzed by changing the mean of F to 1000 kN, keeping coefficient of variation = 0.2. The mean factor of safety is 4.9, which is much higher. The exact reliability index and probability of failure are 3.65 and 1.3×10^{-4}, respectively. The probability of failure from a third-order expansion is 5.4×10^{-4} (direct simulation is 1.5×10^{-4}). This discrepancy cannot be removed by increasing the number of collocation points through random sampling. A fourth-order expansion is needed to match the probability of failure from direct simulation [Fig. 3(bottom)]. The Appendix shows a simple and efficient 2-term recurrence method for obtaining Hermite polynomials of any order in the case of $n = 2$. To our knowledge, this proposal appears to be original. It is important to note that the performance function is evaluated only 25 times in the Hermite expansion method (both third- and fourth-order). Another significant advantage is that an entire vector output can be expressed using the same template expansion (e.g. Eq. 9) with different numerical coefficients.

Conclusions

The efficient propagation of input uncertainties through a physical model to arrive at a vector of output uncertainties is an outstanding challenge. Although this problem can be solved using Monte Carlo simulation, it is well recognized that the computational efforts required can be astronomical for sufficiently complex models that only admit numerical solutions. This paper explores the application of the multi-dimensional Hermite polynomials to reduce an unknown random output vector into a significantly simpler unknown vector of numbers. These unknown numbers can be evaluated using a Galerkin or collocation method. The latter method has the important practical advantage of allowing existing deterministic numerical codes to be used as "black boxes". A simple laterally loaded pile example involving two input random variables demonstrated that a third- or fourth-order Hermite expansion is adequate to reproduce probabilities of failure between 10^{-3} and 10^{-4}. Only 25 runs are needed to produce the same accuracy as 500,000 Monte Carlo simulation runs. Although the multi-dimensional Hermite polynomials appear to be extremely efficient, it suffers from the proliferation of expansion terms at higher dimension and order. It is possible that the order can be kept at four or below for most practical problems, but the number of the random dimension cannot be controlled. For a problem involving 5 input random variables, a third-order expansion would involve 56 terms. A simple and efficient 2-term recurrence method for obtaining Hermite

polynomials of any order in the case of $n = 2$ is proposed. To our knowledge, this proposal appears to be original. However, there appears to be no simple method to manage the proliferation of Hermite expansion terms for higher random dimensions.

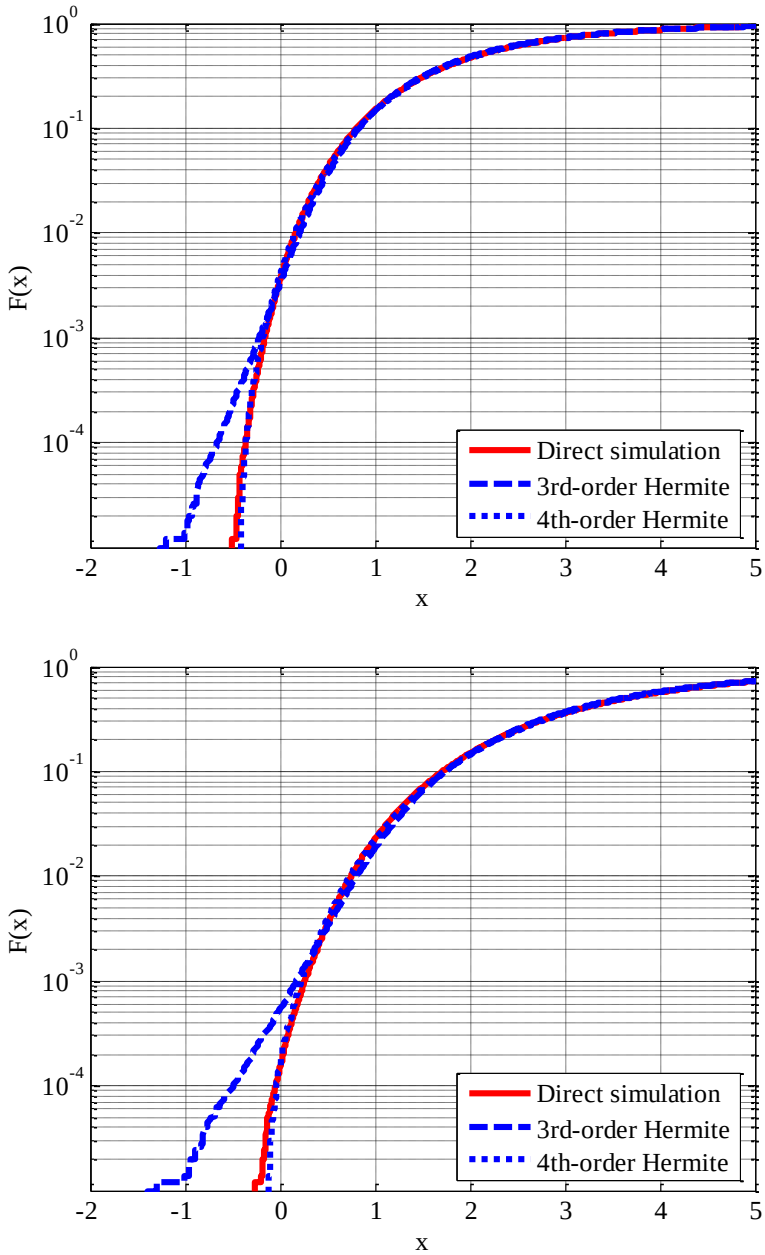
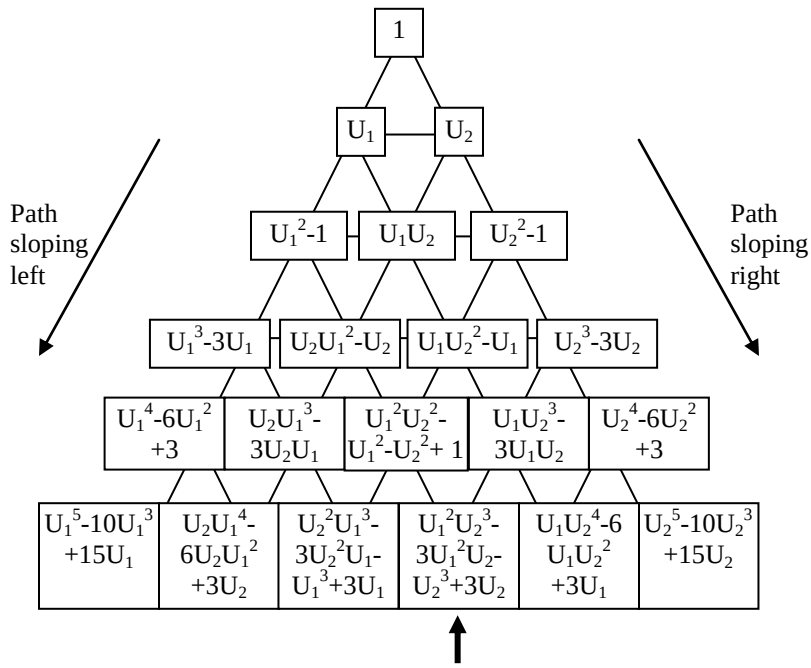


Figure 3. Cumulative distribution function of performance function for mean factor of safety = 3.3 (top) and 4.9 (bottom).

Appendix



For n = 2, Hermite polynomials of any order can be generated using the above Pascal triangle. The underlying rules are:

1. Apply the recursion equation $H_{k+1}(U_1,U_2) = U_2\, H_k(U_1,U_2) – kH_{k-1}(U_1,U_2)$ for paths sloping to the right. The index k is zero at the edge of the triangle.
2. Apply the recursion equation $H_{k+1}(U_1,U_2) = U_1\, H_k(U_1,U_2) – kH_{k-1}(U_1,U_2)$ for paths sloping to the left. The index k is zero at the edge of the triangle.

Note that the recursion formulae are based on the 1D Hermite recursion equation given in last row of Eq. (1).

As an example, consider the Hermite polynomial highlighted by the arrow. If we consider the path sloping to the right, the recursion equation is:

$$H_{k+1}(U_1,U_2) = U_2\, H_k(U_1,U_2) – kH_{k-1}(U_1,U_2)$$

and k+1 = 3 because it is four nodes away from the edge of the triangle (edge node corresponds to k = 0). Substituting the appropriate Hermite polynomials two nodes above current node (along the same sloping path), we have:

$$H_{k+1}(U_1,U_2) = U_2\,(U_1^2U_2^2-U_1^2-U_2^2+ 1) – 2(U_2U_1^2-U_2) = U_1^2U_2^3-3U_1^2U_2-U_2^3+3U_2$$

The same answer is obtained by using the path sloping left. In this case, k+1 = 2 and the recursion equation is:

$$H_{k+1}(U_1,U_2) = U_1(U_1U_2^3- 3U_1U_2) - (U_2^3-3U_2) = U_1^2U_2^3-3U_1^2U_2-U_2^3+3U_2$$

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